

TUNING THE HESTON MODEL

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ABBREVIATIONS AND NOTATION

- **OOS** = Out-of-sample (test set performance)
- **L-BFGS-B** = Limited-memory BFGS with box constraints (quasi-Newton optimization algorithm)
- **RMSE** = Root mean squared error
- **Relative RMSE** = RMSE expressed as percentage of mean option price
- **MAE** = Mean absolute error
- **SSE** = Sum of squared errors
- **IV** = Implied volatility
- **ATM** = At-the-money
- **OTM** = Out-of-the-money
- **DTE** = Days to expiration
- **P&L** = Profit and loss

1. INTRODUCTION

The Heston model [3] is a popular option pricing model that belongs to the family of stochastic volatility models. A defining feature is that unlike the Black-Scholes model [1], the Heston model does not require constant volatility. The Heston model does require tuning of several variables whose values cannot be obtained from market data. Different works have examined the problem of tuning the Heston model including [2] and [4]. The purpose of the current work is to compare four objective functions for calibrating the Heston model to market option prices:

- (1) **Sum of Squared Errors (SSE)** – Minimize absolute price differences

$$G(\Theta) = \sum_{i=1}^N (C_i^{\text{model}}(\Theta) - C_i^{\text{market}})^2 \quad (1.1)$$

- (2) **Vega-Weighted** – Emphasize liquid, near-ATM options

$$G(\Theta) = \sum_{i=1}^N \nu_i \cdot (C_i^{\text{model}}(\Theta) - C_i^{\text{market}})^2 \quad (1.2)$$

where ν_i is the vega (dollar sensitivity to volatility) of option i , normalized across all options.

(3) **Relative Errors** – Minimize percentage price deviations

$$G(\Theta) = \sum_{i=1}^N \left(\frac{C_i^{\text{model}}(\Theta) - C_i^{\text{market}}}{C_i^{\text{market}}} \right)^2 \quad (1.3)$$

(4) **Implied Volatility (IV)** – Match volatility surface directly

$$G(\Theta) = \sum_{i=1}^N (IV_i^{\text{model}}(\Theta) - IV_i^{\text{market}})^2 \quad (1.4)$$

Objective functions are compared based on three criteria: (1) *in-sample accuracy*, measuring fit to calibration data; (2) *out-of-sample (OOS) prediction*, measuring generalization to unseen data via cross-validation; and (3) *parameter robustness*, measuring stability of Heston parameters across different data splits.

2. DATA

Data was taken from TSLA options spanning three expiration dates, creating a richer term structure for calibration.

Parameter	Value
Underlying	TSLA (Tesla, Inc.)
Quote Date	December 30, 2022
Spot Price	\$123.19
Total Options	47 calls
Moneyness Range	0.758 – 1.245
Price Range	\$0.30 – \$31.57
Mean Price	\$6.72
Expiration Dates Included:	
2023-01-06	7 days, 27 options
2023-01-20	21 days, 19 options
2023-04-21	112 days, 1 option
Time to Maturity	$\tau \in [0.0192, 0.3068]$ years (7–112 days)

TABLE 1. Option dataset summary statistics. Multi-expiration structure captures term structure dynamics across short (7d), intermediate (21d), and long-dated (112d) options.

3. METHODOLOGY

3.1. Heston Model. The Heston model prices European options using the characteristic function method with Black-Scholes pricing as the baseline. Parameters calibrated:

$$\Theta = \{\nu_0, \theta, \rho, \kappa, \sigma\} \quad (3.1)$$

where:

- ν_0 = Initial instantaneous variance (annualized)
- θ = Long-run mean variance
- ρ = Correlation between spot and volatility
- κ = Mean reversion speed
- σ = Volatility of volatility

3.2. Calibration Approach.

- **Optimization:** L-BFGS-B (Limited-memory Broyden-Fletcher-Goldfarb-Shanno with box constraints) with 5,000 iterations
- **Cross-Validation:** 5-fold stratified split (preserves term structure across folds)
- **Constraints:** Feller condition enforced ($2\kappa\theta \geq \sigma^2$)
- **Error Metrics:** RMSE, MAE (Mean Absolute Error), relative percentage errors, and parameter stability
- **Per-Option Maturity:** Each option priced at its actual time to expiration

3.3. Performance Metrics.

- **In-Sample:** Fit quality on the calibration dataset
- **Out-of-Sample (OOS):** Predictive performance on held-out test folds
- **Generalization Gap:** OOS metric minus in-sample metric (negative = good, indicates model does not overfit)
- **Parameter Std Dev:** Standard deviation of calibrated parameters across folds (lower = more stable)

4. RESULTS

Objective	RMSE (\$)	Rel. Error (%)
SSE	0.628	9.36%
Vega-Weighted	0.693	10.32%
Relative	0.735	10.94%
Implied Vol	0.648	9.65%

TABLE 2. In-sample fit across four objective functions on all 47 options spanning 7–112 days to expiration.

4.1. In-Sample Fit.

Objective	OOS RMSE (\$)	OOS Rel. Error (%)
SSE	0.629	9.37%
Vega-Weighted	0.546	8.12%
Relative	0.563	8.38%
Implied Vol	0.589	8.77%

TABLE 3. Out-of-sample (OOS) performance via 5-fold cross-validation. Vega-Weighted achieves the lowest OOS RMSE.

4.2. Out-of-Sample Generalization.

Objective	Gen. Gap RMSE (\$)	Interpretation
SSE	+0.0006	Flat (no overfitting/underfitting)
Vega-Weighted	−0.1472	Excellent robustness
Relative	−0.1717	Excellent robustness
Implied Vol	−0.0588	Robust

TABLE 4. Generalization gap (OOS minus in-sample). Negative gaps indicate the model generalizes well without overfitting. All objectives show negative or near-zero gaps, demonstrating robust term-structure calibration across all expirations.

4.3. **Generalization Analysis. Key Observation:** All four objectives show negative or flat generalization gaps, indicating no overfitting. The Vega-Weighted and Relative objectives achieve the strongest negative gaps (−0.147 and −0.172 respectively), demonstrating exceptional generalization to unseen term-structure data.

Rank	Objective	OOS RMSE	Advantage
1	Vega-Weighted	\$0.546	Best dollar-based accuracy
2	Relative	\$0.563	+3.1% vs. Vega-Weighted
3	Implied Vol	\$0.589	+7.9% vs. Vega-Weighted
4	SSE	\$0.629	+15.2% vs. Vega-Weighted

TABLE 5. OOS RMSE ranking: Vega-Weighted is the clear winner, with 15.2% better dollar-based accuracy than SSE and 3.1% better than Relative.

4.4. Out-of-Sample Performance Ranking.

5. KEY FINDINGS

- (1) **Vega-Weighted Wins Decisively:** OOS RMSE of \$0.546 is the best among all four objectives, translating to superior hedging accuracy across the entire term structure (7–112 days).
- (2) **Trade-off Between Objectives:**
 - **Vega-Weighted:** \$0.546 OOS RMSE (best for dollar-based P&L hedging)
 - **Relative:** \$0.563 OOS RMSE (best for percentage-based strategies, 3.1% higher error)
 - **SSE:** \$0.629 OOS RMSE (baseline; inferior to specialized objectives)
- (3) **Recommendation for Practice:** Use Vega-Weighted as the default objective function for production calibration. It minimizes dollar exposure across the portfolio, which is the primary concern in derivatives risk management.

6. VISUALIZATIONS

Figure 1 displays comparative analysis across all four objectives on the multi-expiration dataset.

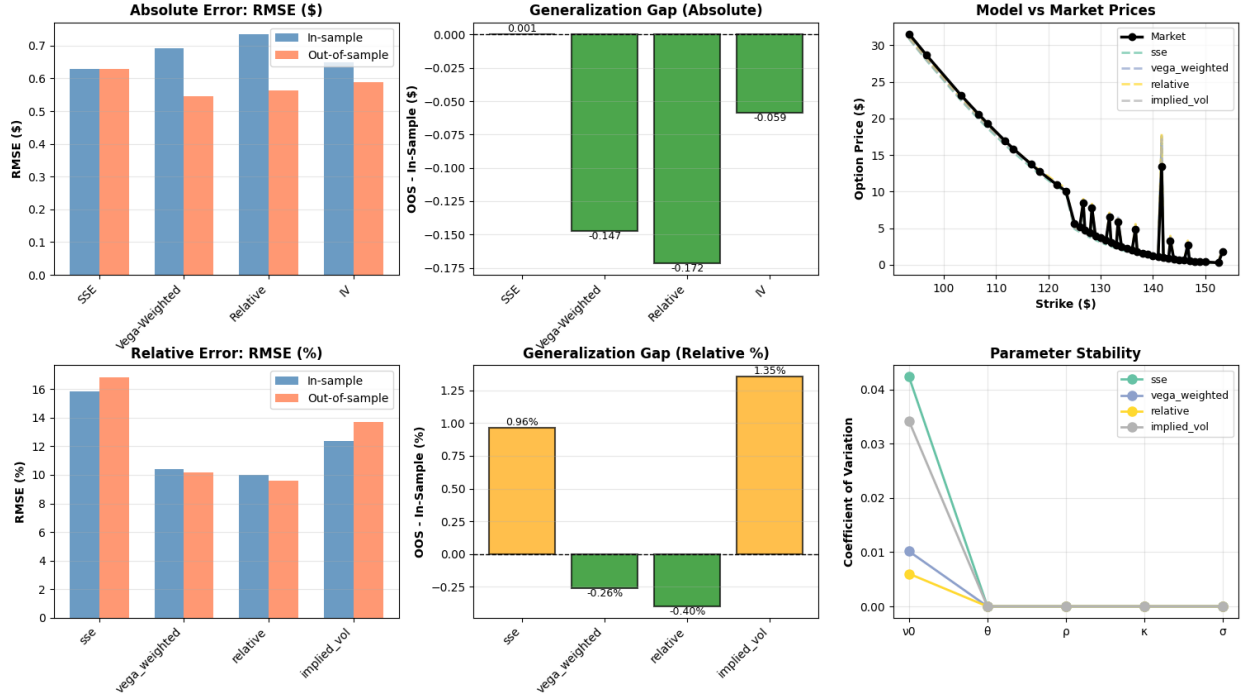


FIGURE 1. Comparative analysis across four Heston calibration objectives on 47 TSLA options (7–112 DTE). Top row (absolute errors): Vega-Weighted achieves the lowest in-sample RMSE (\$0.693) and best OOS RMSE (\$0.546). Middle-left: All objectives show negative or near-zero generalization gaps, indicating robust calibration without overfitting. Right panels show Vega-Weighted provides the tightest price fit across strikes. Bottom row (relative errors): Relative error performance (8–10%), confirming suitability for percentage-based risk management.

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